

# Infinitely Many Integer Solutions for Five Related Diophantine Equations

## Abstract

We prove that each of the five equations

$$\begin{aligned} y^2 + x^3y + z^2 - 2 &= 0, & y^2 + x^3y + z^2 + z - 1 &= 0, \\ y^2 + x^3y + z^2 + z + 1 &= 0, & y^2 + x^3y + y + z^2 + 1 &= 0, \\ y^2 + x^3y + z^2 + 1 &= 0 \end{aligned}$$

has infinitely many integer solutions. The proof is constructive: explicit Pell-type recurrences produce infinitely many distinct values of the parameter  $x$ . A tangent identity on the surface of sums of two squares supplies rational two-square representations, and the classical two-squares theorem converts those rational representations into integral ones.

**Scope of the result.** The paper proves infinitude of integer solutions for the five displayed equations. It does not claim to classify all integer solutions. Each construction below gives an infinite subfamily of solutions, distinguished by infinitely many different values of  $x$ .

## 1 Two elementary tools

We shall use the classical two-squares theorem. A proof is included for self-containment.

**Theorem 1.1** (Fermat's two-squares theorem). *A positive integer  $m$  is a sum of two integer squares if and only if every prime  $p \equiv 3 \pmod{4}$  occurs to an even exponent in the prime factorization of  $m$ .*

*Proof.* We use the Gaussian integers  $\mathbb{Z}[i]$ . For  $\alpha = r + si \in \mathbb{Z}[i]$ , put

$$N(\alpha) = \alpha\bar{\alpha} = r^2 + s^2.$$

This norm is multiplicative, and its units are exactly the elements of norm 1, namely  $\pm 1, \pm i$ .

First recall that  $\mathbb{Z}[i]$  is Euclidean. If  $\alpha, \beta \in \mathbb{Z}[i]$  and  $\beta \neq 0$ , choose a Gaussian integer  $q$  whose real and imaginary parts are nearest integers to those of  $\alpha/\beta$ . Put  $r = \alpha - q\beta$ . Then

$$N(r) = N(\beta) \left| \frac{\alpha}{\beta} - q \right|^2 \leq \frac{1}{2} N(\beta) < N(\beta).$$

Thus Euclidean division is available. Hence every ideal of  $\mathbb{Z}[i]$  is principal: if  $I \neq 0$  is an ideal, choose  $\delta \in I \setminus \{0\}$  with minimal norm. For any  $\alpha \in I$ , write  $\alpha = q\delta + r$ , with either  $r = 0$  or  $N(r) < N(\delta)$ . Since  $r \in I$ , minimality forces  $r = 0$ , so  $I = (\delta)$ . Therefore  $\mathbb{Z}[i]$  is a principal ideal domain.

In a principal ideal domain, every irreducible element is prime. Indeed, if  $\pi$  is irreducible and  $\pi \mid \alpha\beta$  but  $\pi \nmid \alpha$ , then  $(\pi, \alpha) = (1)$ , so there exist  $r, s \in \mathbb{Z}[i]$  with  $r\pi + s\alpha = 1$ . Multiplying by  $\beta$  shows  $\pi \mid \beta$ . Existence of factorizations follows by norm descent: if a nonzero nonunit is not irreducible, write it as a product of two nonunits, each of smaller positive norm, and repeat. The process terminates because positive integers cannot decrease forever. Uniqueness follows by induction on the number of irreducible factors, using the prime property to cancel one irreducible factor at a time. Thus  $\mathbb{Z}[i]$  is a unique factorization domain.

We next classify rational primes in  $\mathbb{Z}[i]$ . The prime 2 ramifies:

$$2 = (1 + i)(1 - i).$$

Let  $p$  be an odd rational prime. If  $p \equiv 3 \pmod{4}$ , then  $-1$  is not a quadratic residue modulo  $p$ . For if  $t^2 \equiv -1 \pmod{p}$ , Fermat's theorem would give

$$1 \equiv t^{p-1} = (t^2)^{(p-1)/2} \equiv (-1)^{(p-1)/2} \equiv -1 \pmod{p},$$

because  $(p-1)/2$  is odd. Here Fermat's theorem follows from the fact that multiplication by  $t$  permutes the nonzero residue classes modulo  $p$ . Thus  $X^2 + 1$  is irreducible over  $\mathbb{F}_p$ , and

$$\mathbb{Z}[i]/(p) \cong \mathbb{F}_p[X]/(X^2 + 1)$$

is a field. Hence  $(p)$  is a prime ideal, equivalently  $p$  is prime in  $\mathbb{Z}[i]$ .

If  $p \equiv 1 \pmod{4}$ , then  $-1$  is a quadratic residue modulo  $p$ . Wilson's theorem for odd primes gives  $(p-1)! \equiv -1 \pmod{p}$ . Also

$$(p-1)! \equiv \prod_{j=1}^{(p-1)/2} j(p-j) \equiv (-1)^{(p-1)/2} \left( \left( \frac{p-1}{2} \right)! \right)^2 \pmod{p}.$$

Since  $p \equiv 1 \pmod{4}$ , the exponent  $(p-1)/2$  is even, and hence

$$\left( \left( \frac{p-1}{2} \right)! \right)^2 \equiv -1 \pmod{p}.$$

Wilson's theorem itself follows by multiplying all nonzero residue classes modulo  $p$ : every class pairs with its inverse, except 1 and  $-1$ , so the product is  $-1$ . Therefore choose  $t$  with  $t^2 \equiv -1 \pmod{p}$ . Then  $p \mid (t+i)(t-i)$  in  $\mathbb{Z}[i]$ , but  $p$  divides neither factor. Thus  $p$  is not prime in  $\mathbb{Z}[i]$ . Since irreducibles are prime in a UFD,  $p$  is not irreducible. Hence  $p = \alpha\beta$  for nonunits  $\alpha, \beta \in \mathbb{Z}[i]$ . Taking norms gives

$$p^2 = N(p) = N(\alpha)N(\beta).$$

The two factors on the right are integers greater than 1. Since  $p$  is a rational prime, the only possibility is  $N(\alpha) = N(\beta) = p$ . Put  $\pi = \alpha$ . Then  $N(\pi) = p$ , and therefore

$$p = \pi\bar{\pi}.$$

Thus a rational prime  $p \equiv 1 \pmod{4}$  splits into two conjugate Gaussian factors, each of norm  $p$ .

Suppose first that  $m = r^2 + s^2$ . Then

$$m = (r + si)(r - si).$$

Let  $p \equiv 3 \pmod{4}$ . As just shown,  $p$  is prime in  $\mathbb{Z}[i]$ , and it is associated to its conjugate. Therefore its exponent in the product  $(r + si)(r - si)$  is twice its exponent in  $r + si$ , hence is even. Thus every prime  $p \equiv 3 \pmod{4}$  occurs to an even exponent in  $m$ .

Conversely, suppose every prime  $p \equiv 3 \pmod{4}$  occurs to an even exponent in  $m$ . Write the prime factorization of  $m$  in  $\mathbb{Z}$ . For each prime  $p \equiv 1 \pmod{4}$ , choose  $\pi_p \in \mathbb{Z}[i]$  with  $N(\pi_p) = p$ . For  $p = 2$ , use  $N(1 + i) = 2$ . For each prime  $q \equiv 3 \pmod{4}$ , the exponent of  $q$  in  $m$  is even, say  $2e_q$ , and  $N(q^{e_q}) = q^{2e_q}$ . Hence the Gaussian integer

$$\alpha = (1 + i)^{v_2(m)} \prod_{p \equiv 1 \pmod{4}} \pi_p^{v_p(m)} \prod_{q \equiv 3 \pmod{4}} q^{v_q(m)/2}$$

has norm  $m$ . Writing  $\alpha = r + si$ , we obtain

$$m = N(\alpha) = r^2 + s^2.$$

This proves the theorem. □

**Lemma 1.2** (Rational-to-integer two-square lemma). *Let  $m$  be a positive integer. If*

$$m = A^2 + B^2$$

*for some  $A, B \in \mathbb{Q}$ , then*

$$m = r^2 + s^2$$

*for some  $r, s \in \mathbb{Z}$ .*

*Proof.* Choose a positive integer  $q$  such that  $qA, qB \in \mathbb{Z}$ . Then

$$mq^2 = (qA)^2 + (qB)^2$$

is a sum of two integer squares. By Theorem 1.1, every prime  $p \equiv 3 \pmod{4}$  occurs to an even exponent in  $mq^2$ . Since  $q^2$  contributes only even exponents, every prime  $p \equiv 3 \pmod{4}$  already occurs to an even exponent in  $m$ . Applying Theorem 1.1 again,  $m$  is a sum of two integer squares. □

We shall also use the Brahmagupta-Fibonacci identity

$$(r^2 + s^2)(p^2 + q^2) = (rp - sq)^2 + (rq + sp)^2. \tag{1}$$

## 2 A tangent identity for $N^6 + c$

The next lemma is the main source of sums of two squares.

**Lemma 2.1** (Tangent identity). *Let  $c, u, a, b \in \mathbb{Z}$ , and put*

$$S = u^3 + c.$$

*Assume that  $S \neq 0$  and*

$$a^2 + b^2 = S.$$

*If  $N, R \in \mathbb{Z}$  satisfy*

$$R^2 = 4SN^2 - u(u^3 - 8c), \tag{2}$$

*then  $N^6 + c$  is a sum of two rational squares.*

*Proof.* The hypotheses imply  $S = a^2 + b^2 > 0$ , so the divisions by  $2S$  below are legitimate. Set

$$L = 3u^2N^2 - u^3 + 2c.$$

We first verify the polynomial identity

$$4S(N^6 + c) - L^2 = (N^2 - u)^2\{4SN^2 - u(u^3 - 8c)\}. \quad (3)$$

Put  $T = N^2$ . The left-hand side of (3) equals

$$\begin{aligned} & 4(u^3 + c)(T^3 + c) - (3u^2T - u^3 + 2c)^2 \\ &= 4(u^3 + c)T^3 - 9u^4T^2 + (6u^5 - 12u^2c)T - u^6 + 8cu^3. \end{aligned}$$

The right-hand side equals

$$\begin{aligned} & (T - u)^2\{4(u^3 + c)T - u(u^3 - 8c)\} \\ &= (T^2 - 2uT + u^2)(4(u^3 + c)T - u^4 + 8uc) \\ &= 4(u^3 + c)T^3 - 9u^4T^2 + (6u^5 - 12u^2c)T - u^6 + 8cu^3. \end{aligned}$$

Thus (3) is proved.

Using (2), identity (3) becomes

$$4S(N^6 + c) - L^2 = (N^2 - u)^2R^2.$$

Define rational numbers

$$A = \frac{aL - b(N^2 - u)R}{2S}, \quad B = \frac{bL + a(N^2 - u)R}{2S}.$$

Since  $S = a^2 + b^2$ , the mixed terms cancel and give

$$\begin{aligned} A^2 + B^2 &= \frac{L^2 + (N^2 - u)^2R^2}{4S} \\ &= \frac{4S(N^6 + c)}{4S} = N^6 + c. \end{aligned}$$

Thus  $N^6 + c$  is a sum of two rational squares. □

The Pell recurrences used below are packaged in the following elementary form.

**Lemma 2.2** (Pell propagation). *Let  $D$  be a positive nonsquare integer, and let  $K \in \mathbb{Z}$ . Suppose  $X_0, N_0, P, Q$  are positive integers satisfying*

$$X_0^2 - DN_0^2 = K, \quad P^2 - DQ^2 = 1.$$

*Define*

$$X_{k+1} = PX_k + DQN_k, \quad N_{k+1} = QX_k + PN_k \quad (k \geq 0).$$

*Then*

$$X_k^2 - DN_k^2 = K$$

*for every  $k \geq 0$ . Moreover, the positive integers  $N_k$  are pairwise distinct. If  $P$  is odd and  $Q$  is even, then the parities of  $X_k$  and  $N_k$  are independent of  $k$ .*

*Proof.* Equivalently,

$$X_k + N_k\sqrt{D} = (X_0 + N_0\sqrt{D})(P + Q\sqrt{D})^k.$$

Taking norms gives  $X_k^2 - DN_k^2 = K$ . Since  $P^2 = 1 + DQ^2 > 1$ , we have  $P > 1$ . Positivity is clear by induction, and

$$N_{k+1} = QX_k + PN_k > N_k.$$

Thus the positive integers  $N_k$  are pairwise distinct. Finally, if  $P$  is odd and  $Q$  is even, then

$$X_{k+1} = PX_k + DQN_k \equiv X_k \pmod{2}, \quad N_{k+1} = QX_k + PN_k \equiv N_k \pmod{2}.$$

□

### 3 Four infinite families of two-square values

**Proposition 3.1.** *There are infinitely many positive even integers  $N$  for which  $N^6 + 8$  is a sum of two integer squares.*

*Proof.* Apply Lemma 2.1 with

$$c = 8, \quad u = 8, \quad a = 6, \quad b = 22.$$

Indeed,

$$6^2 + 22^2 = 520 = 8^3 + 8.$$

For these constants, condition (2) is

$$R^2 = 2080N^2 - 3584 = 16(130N^2 - 224).$$

Thus it is enough to solve

$$X^2 - 130N^2 = -224, \tag{4}$$

and then put  $R = 4X$ . The initial solution and a Pell unit are

$$136^2 - 130 \cdot 12^2 = -224, \quad 6499^2 - 130 \cdot 570^2 = 1.$$

Lemma 2.2, with

$$D = 130, \quad K = -224, \quad X_0 = 136, \quad N_0 = 12, \quad P = 6499, \quad Q = 570,$$

produces infinitely many positive integers  $N_k$  solving (4). Since  $N_0$  is even,  $P$  is odd, and  $Q$  is even, the parity assertion in Lemma 2.2 shows that every  $N_k$  is even. Lemma 2.1 gives rational representations of  $N_k^6 + 8$  as sums of two squares, and Lemma 1.2 converts them to integer representations. □

**Proposition 3.2.** *There are infinitely many positive even integers  $N$  for which  $N^6 + 5$  is a sum of two integer squares.*

*Proof.* Apply Lemma 2.1 with

$$c = 5, \quad u = 2, \quad a = 2, \quad b = 3.$$

Here

$$2^2 + 3^2 = 13 = 2^3 + 5.$$

The condition (2) becomes

$$R^2 = 52N^2 + 64 = 4(13N^2 + 16).$$

Thus it is enough to solve

$$X^2 - 13N^2 = 16, \tag{5}$$

and put  $R = 2X$ . We have

$$22^2 - 13 \cdot 6^2 = 16, \quad 649^2 - 13 \cdot 180^2 = 1.$$

Lemma 2.2, with

$$D = 13, \quad K = 16, \quad X_0 = 22, \quad N_0 = 6, \quad P = 649, \quad Q = 180,$$

therefore gives infinitely many positive integers  $N_k$  solving (5). Since  $N_0$  is even,  $P$  is odd, and  $Q$  is even, the same lemma shows that every  $N_k$  is even. Lemma 2.1 and Lemma 1.2 then imply that each  $N_k^6 + 5$  is a sum of two integer squares.  $\square$

**Proposition 3.3.** *There are infinitely many positive even integers  $N$  for which  $N^6 - 3$  is a sum of two integer squares.*

*Proof.* Apply Lemma 2.1 with

$$c = -3, \quad u = 2, \quad a = 1, \quad b = 2.$$

Indeed,

$$1^2 + 2^2 = 5 = 2^3 - 3.$$

The condition (2) becomes

$$R^2 = 20N^2 - 64 = 4(5N^2 - 16).$$

Thus it is enough to solve

$$X^2 - 5N^2 = -16, \tag{6}$$

and put  $R = 2X$ . We have

$$2^2 - 5 \cdot 2^2 = -16, \quad 9^2 - 5 \cdot 4^2 = 1.$$

Lemma 2.2, with

$$D = 5, \quad K = -16, \quad X_0 = 2, \quad N_0 = 2, \quad P = 9, \quad Q = 4,$$

gives infinitely many positive integers  $N_k$  solving (6). Since  $N_0$  is even,  $P$  is odd, and  $Q$  is even, Lemma 2.2 shows that every  $N_k$  is even. Lemma 2.1 and Lemma 1.2 then imply that each  $N_k^6 - 3$  is a sum of two integer squares.  $\square$

**Proposition 3.4.** *There are infinitely many positive odd integers  $N$  for which  $N^6 - 4$  is a sum of two integer squares.*

*Proof.* Apply Lemma 2.1 with

$$c = -4, \quad u = 242, \quad a = 578, \quad b = 3720.$$

Then

$$a^2 + b^2 = 578^2 + 3720^2 = 14172484 = 242^3 - 4.$$

Thus, in the notation of Lemma 2.1,  $S = 14172484 \neq 0$ . The condition (2) becomes

$$R^2 = 56689936N^2 - 3429749840.$$

Since

$$56689936 = 16 \cdot 3543121, \quad 3429749840 = 16 \cdot 214359365,$$

it is enough to solve

$$X^2 - 3543121N^2 = -214359365, \tag{7}$$

and then put  $R = 4X$ . The integer 3543121 is nonsquare, since

$$1882^2 < 3543121 < 1883^2.$$

The following two identities are direct integer equalities:

$$\begin{aligned} 410154078658987088294^2 - 3543121 \cdot 217898400660155259^2 &= -214359365, \\ 53319160674725436041296699414748449^2 \\ - 3543121 \cdot 28326330128303226243386806483560^2 &= 1. \end{aligned}$$

Now apply Lemma 2.2 with

$$\begin{aligned} D &= 3543121, & K &= -214359365, \\ X_0 &= 410154078658987088294, & N_0 &= 217898400660155259, \\ P &= 53319160674725436041296699414748449, \\ Q &= 28326330128303226243386806483560. \end{aligned}$$

This produces infinitely many positive integers  $N_k$  solving (7). The parity is preserved because  $P$  is odd and  $Q$  is even. Since  $N_0$  is odd, every  $N_k$  is odd. Lemma 2.1 gives rational representations of  $N_k^6 - 4$  as sums of two squares, and Lemma 1.2 converts them to integer representations.  $\square$

## 4 Converting two-square values into solutions

We now turn the two-square values above into solutions of the five equations.

**Theorem 4.1.** *Each of the equations*

$$y^2 + x^3y + z^2 - 2 = 0, \tag{E1}$$

$$y^2 + x^3y + z^2 + z - 1 = 0, \tag{E2}$$

$$y^2 + x^3y + z^2 + z + 1 = 0 \tag{E3}$$

*has infinitely many integer solutions.*

*Proof.* For (E1), let  $N$  run through the infinitely many positive even integers supplied by Proposition 3.1. For each such  $N$ , choose integers  $a, b$  with

$$N^6 + 8 = a^2 + b^2.$$

Since  $N$  is even,  $N^6 + 8 \equiv 8 \pmod{16}$ . A square modulo 16 is one of 0, 1, 4, 9, so a representation of a number congruent to 8 (mod 16) forces both  $a$  and  $b$  to be even. Put

$$x = N, \quad y = \frac{a - N^3}{2}, \quad z = \frac{b}{2}.$$

Then  $y, z \in \mathbb{Z}$ , and

$$(2y + x^3)^2 + (2z)^2 = a^2 + b^2 = N^6 + 8 = x^6 + 8.$$

Expanding and dividing by 4 gives

$$y^2 + x^3y + z^2 - 2 = 0.$$

The values  $x = N$  are pairwise distinct, so (E1) has infinitely many integer solutions.

For (E2), let  $N$  run through the infinitely many positive even integers supplied by Proposition 3.2. Choose integers  $a, b$  with

$$N^6 + 5 = a^2 + b^2.$$

Since  $N$  is even,  $N^6 + 5 \equiv 5 \pmod{8}$ . Hence one of  $a, b$  is even and the other is odd. Relabel them, if necessary, so that  $a$  is even and  $b$  is odd. Put

$$x = N, \quad y = \frac{a - N^3}{2}, \quad z = \frac{b - 1}{2}.$$

Then  $y, z \in \mathbb{Z}$ , and

$$(2y + x^3)^2 + (2z + 1)^2 = a^2 + b^2 = N^6 + 5 = x^6 + 5.$$

Expanding, cancelling  $x^6$ , and dividing by 4 gives

$$y^2 + x^3y + z^2 + z - 1 = 0.$$

Again the values  $x = N$  are pairwise distinct, so (E2) has infinitely many integer solutions.

For (E3), use the infinitely many positive even integers  $N$  supplied by Proposition 3.3. Choose integers  $a, b$  with

$$N^6 - 3 = a^2 + b^2.$$

Since  $N$  is even,  $N^6 - 3 \equiv 5 \pmod{8}$ . Thus one of  $a, b$  is even and the other is odd; relabel so that  $a$  is even and  $b$  is odd. Put

$$x = N, \quad y = \frac{a - N^3}{2}, \quad z = \frac{b - 1}{2}.$$

Then  $y, z \in \mathbb{Z}$ , and

$$(2y + x^3)^2 + (2z + 1)^2 = a^2 + b^2 = N^6 - 3 = x^6 - 3,$$

and expanding, cancelling  $x^6$ , and dividing by 4 gives

$$y^2 + x^3y + z^2 + z + 1 = 0.$$

The values  $x = N$  are pairwise distinct, so (E3) has infinitely many integer solutions. □

**Theorem 4.2.** *The equation*

$$y^2 + x^3y + y + z^2 + 1 = 0 \tag{E4}$$

*has infinitely many integer solutions.*



*Proof.* Let  $N$  run through the infinitely many positive even integers supplied by Proposition 3.3. Thus

$$N^6 - 3 = r^2 + s^2$$

for some integers  $r, s$ . Also

$$N^6 + 1 = (N^3)^2 + 1^2.$$

By (1), the product

$$(N^6 - 3)(N^6 + 1)$$

is a sum of two integer squares. Now set

$$x = -N^2.$$

Then

$$\begin{aligned} (x^3 + 1)^2 - 4 &= (1 - N^6)^2 - 4 \\ &= N^{12} - 2N^6 - 3 \\ &= (N^6 - 3)(N^6 + 1). \end{aligned}$$

Therefore there exist integers  $a, b$  such that

$$(x^3 + 1)^2 - 4 = a^2 + b^2.$$

Because  $N$  is even,  $x = -N^2$  is even, so  $x^3 + 1$  is odd and

$$(x^3 + 1)^2 - 4 \equiv 5 \pmod{8}.$$

Thus one of  $a, b$  is odd and the other is even. Relabel, if necessary, so that  $a$  is odd and  $b$  is even. Put

$$y = \frac{a - (x^3 + 1)}{2}, \quad z = \frac{b}{2}.$$

Then  $y, z \in \mathbb{Z}$ , and

$$(2y + x^3 + 1)^2 + (2z)^2 = a^2 + b^2 = (x^3 + 1)^2 - 4.$$

Expanding, cancelling  $(x^3 + 1)^2$ , and dividing by 4 gives

$$y^2 + x^3y + y + z^2 + 1 = 0.$$

The values  $x = -N^2$  are pairwise distinct as  $N$  varies, so (E4) has infinitely many integer solutions.  $\square$

**Theorem 4.3.** *The equation*

$$y^2 + x^3y + z^2 + 1 = 0 \tag{E5}$$

*has infinitely many integer solutions.*

*Proof.* Let  $N$  run through the infinitely many positive odd integers supplied by Proposition 3.4. For each such  $N$ , choose integers  $a, b$  with

$$N^6 - 4 = a^2 + b^2.$$

Since  $N$  is odd,

$$N^6 - 4 \equiv 1 - 4 \equiv 5 \pmod{8}.$$

The quadratic residues modulo 8 are 0, 1, 4. Therefore one of  $a, b$  is odd and the other is even. Relabel them, if necessary, so that  $a$  is odd and  $b$  is even. Put

$$x = N, \quad y = \frac{a - N^3}{2}, \quad z = \frac{b}{2}.$$

These are integers, because  $a$  and  $N^3$  are odd and  $b$  is even. Also

$$(2y + x^3)^2 + (2z)^2 = a^2 + b^2 = N^6 - 4 = x^6 - 4.$$

Expanding the left-hand side gives

$$4y^2 + 4x^3y + x^6 + 4z^2 = x^6 - 4.$$

After cancellation and division by 4, this is

$$y^2 + x^3y + z^2 + 1 = 0.$$

The values  $x = N$  are pairwise distinct, so (E5) has infinitely many integer solutions.  $\square$

Combining Theorems 4.1, 4.2, and 4.3 proves the desired claim for all five equations.

*Remark 4.4.* The construction gives explicit infinite families. The  $N$ 's in Propositions 3.1, 3.2, 3.3, and 3.4 are obtained from the stated Pell recurrences. The proof does not claim to classify all integer solutions; it only supplies infinitely many of them.

## A Exact arithmetic certificate for the large Pell identities

The proof of Proposition 3.4 uses two large Pell-type identities. They are ordinary integer equalities, not heuristic or floating-point checks. The following plain Python 3 script verifies them using exact arbitrary-precision integer arithmetic.

Listing 1: Exact verification of the large Pell norm identities.

```
D = 3543121
C = 214359365

X0 = 410154078658987088294
N0 = 217898400660155259

P = 53319160674725436041296699414748449
Q = 28326330128303226243386806483560

initial_norm = X0*X0 - D*N0*N0
unit_norm = P*P - D*Q*Q

assert initial_norm == -C
assert unit_norm == 1

# Parity data used in the c = -4 construction.
assert X0 % 2 == 0
assert N0 % 2 == 1
assert P % 2 == 1
assert Q % 2 == 0

print("initial_norm =", initial_norm)
print("unit_norm =", unit_norm)
print("parities (X0,N0,P,Q) =", (X0 % 2, N0 % 2, P % 2, Q % 2))
```

The expected output is

```
initial_norm = -214359365
unit_norm = 1
parities (X0,N0,P,Q) = (0, 1, 1, 0)
```

For completeness, the smaller constants used to specialize Lemma 2.1 in Proposition 3.4 may be checked in the same way:

Listing 2: Exact verification of the constants in Proposition 3.4.

```
u = 242
S = u**3 - 4

assert S == 14172484
assert 578**2 + 3720**2 == S
assert 4*S == 16*3543121
assert u*(u**3 + 32) == 16*214359365

print("small constants verified")
```

These scripts are included only as checkable certificates for arithmetic substitutions. The mathematical proof of infinitude is the Pell recurrence and norm argument given in Lemma 2.2 and Proposition 3.4.